

Assignment 3

SQL Aggregate Functions and Operates

Section 1

① The Students Table

Table Students

Student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

SELECT DISTINCT department
FROM Students;

department
IT
HR
Finance

② Average age of students per department

SELECT department, AVG(age) As avg-age
FROM Students
GROUP BY department;

department	Avg-age
IT	20.5
HR	22
Finance	23

③ Department with more than 1 student.

SELECT department, COUNT(*) AS student_count
FROM students

GROUP BY department

HAVING COUNT(*) > 1;

department	student_count
IT	2
HR	2

④ All students whose is aged between 21 and 23

SELECT student_id, name, age, department
FROM students

WHERE age BETWEEN 21 and 23

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

⑤ All students in the IT or HR department who are older than 21

SELECT student_id, name, age, department
FROM students

WHERE department IN ('IT', 'HR')

AND age > 21;

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

② Section 2

The courses Table

Courses

course_id	course_name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

⑥ Total credits per department, only for departments with more than 5 total credits

```
SELECT department, SUM(credits) AS total_credits  
FROM Courses
```

```
GROUP BY department
```

```
HAVING SUM(credits) > 5;
```

department	total_credits
IT	11

⑦ Courses that do not have 4 Credits

```
SELECT course_id, course_name, department,  
credits
```

```
FROM courses
```

```
WHERE credits != 4;
```

Course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

⑥ Total credits per department, only for departments with more than 5 credits

```
SELECT department, sum(credits) As total_credits  
FROM courses
```

⑧ Top 3 ~~by~~ courses by credits in descending order.

```
SELECT course_id, course_name, credits  
FROM courses  
ORDER BY credits DESC  
LIMIT 3;
```

course_id	Course_name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Section 3

⑨ Get the maximum, minimum and average all enrollments

```
SELECT  
    MAX(grade) As max_grade  
    MIN(grade) As min_grade  
    AVG(grade) As avg_grade  
FROM enrollments;
```

~~Enrollments~~

Maximum_grade	Minimum_grade	Average
90	78	84.6

Enrollments

enrollment_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

⑩ Count how many enrollments exist per course

```
SELECT
    course_id,
    count(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

~~Section 4~~

~~⑪ Find total salary and total bonus per department~~

~~SELECT department
SUM(salary)~~

Section 4

Salaries

employee_id	name	department	Salary	Bonus
1	Tom	IT	60000	5000
2	Jerry	HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyke	IT	62000	5500
5	Butch	HR	54000	3500

⑪ find total salary and total bonus per dep.

```
SELECT
    SUM(salary) AS total_salary
    SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

⑫ Show departments whose average salary is above 55 000

```
SELECT department,
    AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

department	avg salary
IT	61000
Finance	70000

⑬ List employees whose salary + bonus is greater than 60 000

```
SELECT employee_id,
    name,
    salary,
    bonus,
    salary + bonus AS total_compensation.
FROM salaries
WHERE salary + bonus > 60000;
```

Section 5

Projects

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000

⑭ Total and average budget per department only include department with average budget above 70 000.

SELECT

department,

SUM(budget) AS total_budget

AVG(budget) AS avg_budget

FROM projects

GROUP BY department

HAVING AVG(budget) > 70000;

department	total-budget	avg total-budget
IT	270 000	135 000
Finance	80 000	80 000

⑮ Projects with budgets between 50 000 and 120 000 excluding Marketing.

SELECT project_id,
project_name,
department,
budget

FROM projects
WHERE budget BETWEEN 50 000 and
120 000
AND department < > 'Marketing';